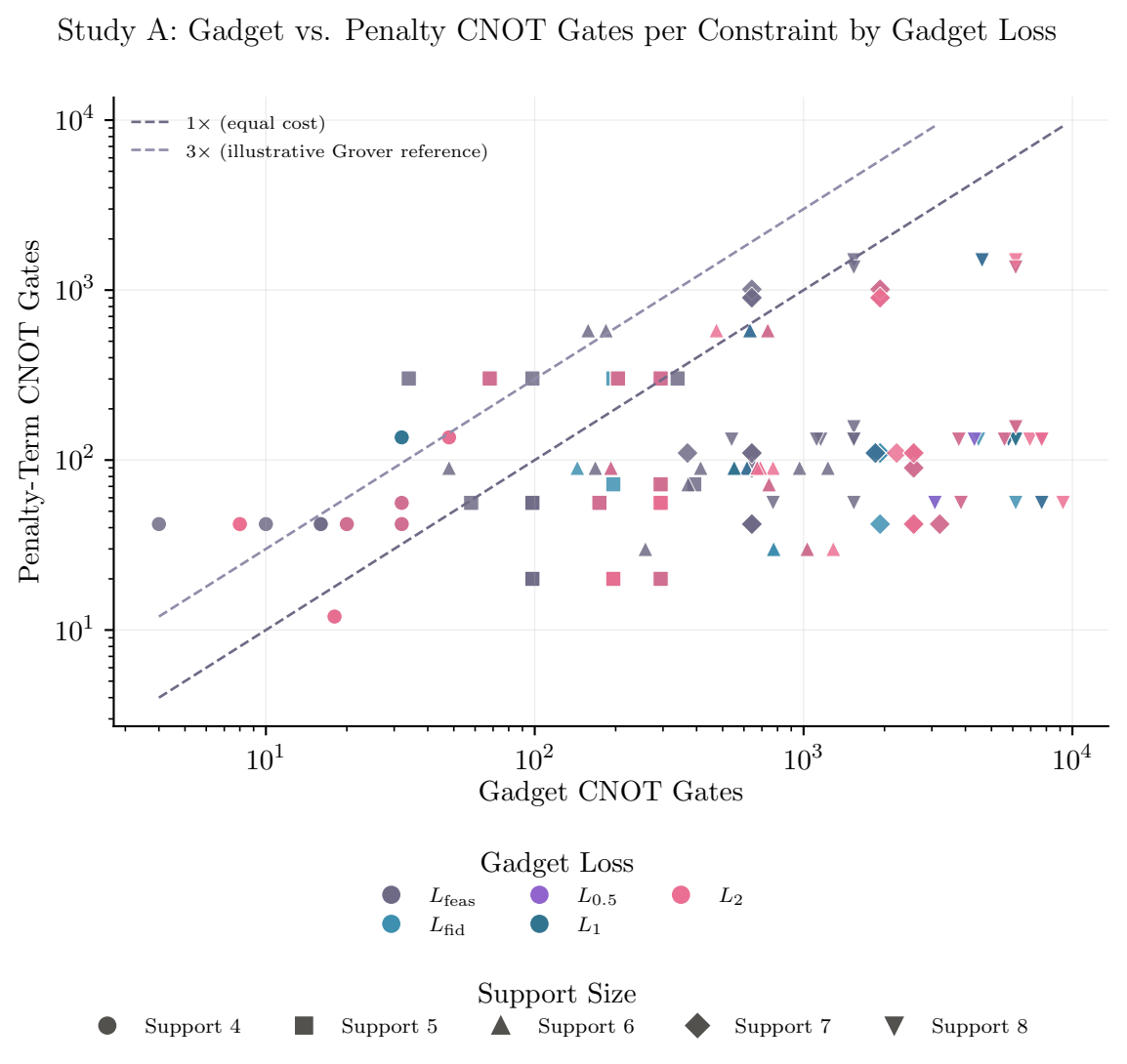

Study A: Gadget vs. Penalty CNOT Gates per Constraint by Gadget Loss
1× (equal cost)
3× (illustrative Grover reference)
Penalty-Term CNOT Gates
Gadget CNOT Gates
Gadget Loss
$L_{\text{feas}}$
$L_{0.5}$
$L_2$
$L_{\text{fid}}$
$L_1$
Support Size
Support 4
Support 5
Support 6
Support 7
Support 8